

Education

B.Tech in CSE (Computer Science and Engineering) Indraprastha Institute of Information Technology Delhi (IIIT-Delhi)	2022 – 2026
Class XII – CBSE Bhiwani Public School, Bhiwani, Haryana	2020 – 2021
Class X – CBSE Halwasiya Vidya Vihar, Bhiwani, Haryana	2018 – 2019

Experience

Software Intern — LLM Training Lead <i>DataPattern, Chennai</i>	May 2025 – July 2025
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- Led a team project to design and train a private, on-premise Large Language Model on proprietary company data.
- Enabled the organisation to operate a fully internal AI system with zero dependence on external cloud-based AI services.
- Project undertaken under strict confidentiality protocols for a client operating in critical national infrastructure domains including defence and space.
- Gained hands-on experience in LLM fine-tuning, data pipelines, and secure model deployment.

Skills

Languages	Python, Java, Kotlin, C, C++, SQL, JavaScript, HTML, CSS, Bash
Frameworks	React.js, FastAPI, LibGDX, JavaFX, Firebase, SQLAlchemy, Pydantic, Uvicorn, Vite, React Router, Web Speech API
Mobile Dev	Android Studio, Firebase Auth, Firebase Realtime DB, Firebase Cloud Messaging (FCM), Material Design
Databases	PostgreSQL, MySQL, Firebase Realtime DB, SQLite
AI / ML	LLM Fine-tuning, Collaborative Filtering, Nvidia NIM API (LLaMA 3.3-70b), NLP, Prompt Engineering, Recommendation Systems
Security	JWT, Argon2, SSL/TLS, RBAC, OpenSSL, PKI, OTP Auth, XSS/SQL Injection Defence, Session Management
DevOps & Tools	Git, GitHub, Docker, Postman, Nginx, Linux (Ubuntu/Kali/Arch), VS Code, IntelliJ, PyCharm, Figma
Concepts	OOP, Data Structures & Algorithms, Operating Systems, REST APIs, System Design, MVC Architecture, Multithreading

Projects

CriticalLens — AI-Powered Adaptive Learning Platform <i>React.js, FastAPI, Nvidia NIM API (LLaMA 3.3-70b), Web Speech API</i>	<i>Team of 2</i>
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- Built an adaptive learning platform to combat passive AI consumption among students, grounded in research from MIT Media Lab and Aalto University.
- Implemented two distinct learning modes (Learn Mode and Revise Mode) with adaptive difficulty engine, confidence tracking, and metacognitive feedback.
- Integrated document upload (PDF/Word/Image), voice input, and a multi-key API rotation system for uninterrupted availability.

Hybrid Collaborative Filtering + LLM Fine-tuning <i>Python, Machine Learning, LLM Fine-tuning</i>	<i>Solo</i>
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- Designed a hybrid recommendation system combining collaborative filtering with a fine-tuned LLM on the Amazon Video Games dataset.
- Focused on personalised recommendations through learned user-item interaction patterns.

SecureJob — Secure Job Search Platform
React 18, FastAPI, PostgreSQL, Nginx, OpenSSL, JWT, Argon2

Team

- Built a full-stack job search platform with production-grade security — SSL/TLS, JWT, Argon2 hashing, RBAC, and Nginx reverse proxy.
- Secured against SQL injection, XSS, and session hijacking across a fully documented REST API.

Online Retail Store — DBMS Project
SQL, Python, MySQL

Team

- Designed a relational database for a multi-role retail platform with automated triggers, query optimisation, and role-based access for Admin, Seller, and Customer workflows.

Also worked on:

- **WhatsApp Clone** — Real-time messaging Android app using Firebase, Java/Kotlin with push notifications and media sharing.
- **PhonePe Clone** — Android payment app with simulated payment gateway, user authentication, and transaction history.
- **Angry Birds Game** — Java + LibGDX game with projectile physics, collision detection, and cross-platform support.
- **Stick Hero Game** — JavaFX-based game demonstrating core game development mechanics.
- **Secure Social Media Marketplace** — End-to-end encrypted platform with P2P marketplace, OTP auth, PKI encryption, and admin moderation dashboard.
- **Simple Assembler & Simulator** — Python-based assembler and simulator demonstrating low-level systems programming.
- **Simple Shell** — C-based shell with command parsing, process management, and memory-efficient child process execution.
- **AI Recipe Recommender** — ML-based recipe suggestion system with NLP query support and dietary preference filtering.

Positions of Responsibility

Events Team & PR Lead — Cyfuse (Tech Club, IIIT Delhi)	Jan 2024 – Present
Core Member & Head Speaker — Cyfuse	Oct 2024 – Present
Operations Team — Odyssey 2024 (Annual Tech Fest)	Jan 2024
Security Head	Jan 2025

Achievements & Activities

- Completed 25 km Cycle Marathon (2019)
- Deadlift Champion — inter-college competition
- District Olympiad Winner
- Gaming Champion — NCT of Delhi
- Volunteer — Medical awareness and education drives in underprivileged communities

Relevant Coursework

Core	Introduction to Programming · Data Structures & Algorithms · Advanced Programming (OOP in Java) · Computer Organisation · Discrete Mathematics · Analysis & Design of Algorithms · Operating Systems · Computer Networks · Database Systems
AI / ML	Machine Learning · Deep Learning · Artificial Intelligence · Collaborative Filtering · Natural Language Processing
Security	Foundations of Computer Security · Ethical Hacking · Network Security · Security Engineering
Systems & Software	Software Engineering · Mobile Computing · Compilers · Theory of Computation · Distributed Systems Security
Specialised	Human-AI Interaction · Computer Vision · Digital & Cyber Forensics · GPU Computing

Interests

Artificial Intelligence & Machine Learning · Systems Programming · Mobile & Android Development · Cybersecurity · Open Source Contribution · Human-AI Interaction Design · Music Production & Listening · Gaming · Fitness & Strength Training